

$$\int_0^{\frac{\pi}{4}} \frac{\cos^3 2x}{1+\sin^2 2x} dx$$

$$u:=2x \quad du=2dx$$

$$\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{\cos^2u\cos u}{1+\sin^2u}du$$

$$\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{(1-\sin^2u)\cos u}{1+\sin^2u}du$$

$$w:=\sin u \quad dw=\cos u du$$

$$\frac{1}{2}\int_0^1\frac{1-w^2}{1+w^2}dw$$

$$x^2+1\Bigg|\frac{-1}{-x^2+1}\\\frac{-(-x^2-1)}{2}$$

$$\frac{1}{2}\int_0^1\Big(-1+\frac{2}{1+w^2}\Big)dw$$

$$\frac{1}{2}(-w+2\arctan w)|_0^1$$

$$\frac{1}{2}\Big(-1+\frac{\pi}{2}\Big)$$

$$-\frac{1}{2}+\frac{\pi}{4}$$

$$\frac{\pi}{4}-\frac{1}{2}$$